Daily Logic Drill #6

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Evaluating Compound Claims · AND/OR Combinations · The "Which Statements Are True" Format · Quantifier Mixing.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition (≤15 s each)

Evaluate each statement independently before combining. Pure recognition.

1. I. "2 is prime." II. "2 is odd." Which are true?
2. I. "4 is a perfect square." II. "4 is prime." Which are true?
3. I. "9 is odd." II. "9 is prime." Which are true?
4. I. "Every square is a rectangle." II. "Every rectangle is a square." Which are true?
5. I. "6 is a multiple of 2." II. "6 is a multiple of 3." Which are true?
6. I. "5 is even." II. "5 is prime." Which are true?
7. I. "0 is positive." II. "0 is negative." Which are true?
8. True or false: "If both statements I and II in a pair are false, the correct choice is 'neither', not 'I and II'."
9. I. "$n = 2$ is a counterexample to 'all primes are odd'." II. "$n = 9$ is a counterexample to 'all odd numbers are prime'." Which are true?
10. I. "1 is a prime number." II. "1 is a perfect square." Which are true?

Section B Manipulation Drills (≤50 s each)

Three statements at once — evaluate each on its own merits before combining.

1. I. "$n = 4$ is a multiple of 4." II. "$n = 4$ is a multiple of 8." III. "$n = 4$ is a multiple of 2." Which are true?
2. I. "n even $\Rightarrow n^2$ even" is true. II. Its converse is also true. III. Together, I and II mean "n even $\iff n^2$ even." Which are true?

3. I. "$\sqrt{2}$ is irrational." II. "$\sqrt{2}+\sqrt{2}$ is irrational." III. "$\sqrt{2}-\sqrt{2}$ is irrational." Which are true?

4. I. "Every multiple of 9 is a multiple of 3." II. "Every multiple of 3 is a multiple of 9." III. "'Multiple of 3' is necessary for 'multiple of 9'." Which are true?

5. I. "$2^{11}-1$ is prime." II. "11 is prime." III. "If n is prime, 2^n-1 is prime." Which are true?

6. I. "$4!+1$ is prime." II. "$5!+1$ is prime." III. "$n!+1$ is prime for all n." Which are true?

7. I. "41 is prime." II. "$n=41$ is a counterexample to 'n^2+n+41 is always prime'." III. "$n=40$ is the smallest counterexample to 'n^2+n+41 is always prime'." Which are true?

8. I. "The negation of 'for all x, $P(x)$' is 'for all x, not $P(x)$'." II. "The negation of 'for all x, $P(x)$' is 'there exists x, not $P(x)$'." III. "The negation of 'there exists x, $P(x)$' is 'for all x, not $P(x)$'." Which are true?

9. I. "The converse of a true statement is always true." II. "The contrapositive of a true statement is always true." III. "The negation of a true statement is always true." Which are true?

10. I. "Affirming the consequent is a valid rule." II. "Denying the antecedent is a valid rule." III. "Modus ponens is a valid rule." Which are true?

Section C Substitution & Structure (≤90 s each)

Harder combinations, drawing on facts from across the whole pillar so far.

1. For a positive integer n: I. "n divisible by 4" is necessary for "n divisible by 12." II. "n divisible by 4" is sufficient for "n divisible by 12." III. "n divisible by 4" is necessary and sufficient for "n divisible by 8." Which of I, II, III are true?

A) I only

B) II only

C) III only

D) I and II only

E) I and III only

F) II and III only

G) I, II and III

H) None of them

2. I. "There exists a real x with $x^2<0$." II. "For all reals x, $x^2\geq 0$." III. "The negation of II is I." Which are true?

3. Recall the flawed argument "n^2 multiple of 4, therefore n multiple of 4" (Day 5, B1). I. "This argument commits the fallacy of affirming the consequent." II. "$n=6$ is a valid counterexample to the underlying claim." III. "The converse of the underlying claim ('n mult. of $4 \Rightarrow n^2$ mult. of 4') is what's actually true." Which are true?

4. I. "$n = 1$ is a counterexample to 'every positive integer has a prime factor'." II. "$n = 1$ is a counterexample to 'every integer greater than 1 has a prime factor'." III. "1 is neither prime nor composite." Which are true?

5. I. "The statement 'if $n^2 - n$ is even, then n is even' is false (Day 1, D5)." II. "$n^2 - n$ is even for every integer n." III. "$n(n-1)$ is even for every integer n." Which are true?

6. I. "Every quadratic with integer coefficients that has an integer root has two integer roots." II. "$2x^2 - 3x + 1$ is a counterexample to I." III. "Monic quadratics (leading coefficient 1) are never counterexamples to I." Which are true?

7. I. "$3^5 + 2$ is composite (Day 4, C6)." II. "$3^5 + 2 = 245$." III. "$245 = 5 \times 7^2$." Which are true?

8. For real x: I. "$x > 0$" is sufficient for "$x^2 > 0$." II. "$x > 0$" is necessary for "$x^2 > 0$." III. "$x \neq 0$" is necessary and sufficient for "$x^2 > 0$."

A) I only

B) I and II only

C) I and III only

D) II and III only

E) I, II and III

Section D Challenge (TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. For a positive integer n: I. "n divisible by 6" is necessary and sufficient for "n divisible by 2 and by 3." II. "n^2 divisible by 9" is necessary and sufficient for "n divisible by 3." III. "n^2 divisible by 4" is necessary and sufficient for "n divisible by 4." Which of I, II, III are true?

A) I only

B) II only

C) III only

D) I and II only

E) I and III only

F) II and III only

G) I, II and III

H) None of them

2. Three people, X, Y, Z, each make one statement. X says: "Y is lying." Y says: "Z is lying." Z says: "X and Y are both lying." Given that exactly one of them is telling the truth, determine who.

3. Recall the flawed conclusion from Day 5, D2 (a student proves $\sqrt{2}$ irrational by contradiction, then writes "therefore $\sqrt{2}$ is rational"). I. "The algebraic derivation, up to the final sentence, is entirely correct." II. "The final English conclusion correctly follows from the derived contradiction." III. "A correctly-stated conclusion would be '$\sqrt{2}$ is irrational'." Which are true?

4. Let S be the statement: "If n is a multiple of 12, then n is a multiple of 3 and a multiple of 4." I. S is true. II. S's converse is true. III. "Multiple of 12" is necessary and sufficient for "multiple of 6." Which are true?

5. Statement T: "If $2^n - 1$ is prime, then n is prime" (Day 4, D5). I. T's contrapositive is "if n is not prime, then $2^n - 1$ is not prime." II. T's converse is false, witnessed by $n = 11$. III. "$2^n - 1$ is prime" is necessary and sufficient for "n is prime." Which are true?

"Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised."

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths